

Tutorials - 4 (Vibrations)

1. In seconds pendulum, the amplitude falls to half of its initial value in 150 sec. Calculate Q factor and relaxation time.
2. If the quality factor of an under damped tuning fork of frequency 256Hz is 1000, calculate the time in which its energy is reduced to $1/e$ of the energy in the absence of damping. How many oscillations the tuning fork will make in this time? Also calculate the damping constant and relaxation time.
3. A damped harmonic oscillator has its first amplitude 16 cm and it reduces to 1 cm after 20 oscillations. If the period of oscillation is 6.9 s, calculate the relaxation time.
4. The energy of a piano string of frequency 256 Hz reduces to half of its initial value in 2 s. What is the quality factor of the string?
5. The Q value of a spring loaded with 0.3 kg is 60. It vibrates with a frequency of 2 Hz. Calculate the force constant and mechanical resistance.
6. If the quality factor of an undamped tuning fork of frequency 256 is 103, calculate the time in which its energy is reduced to $(1/e)$ of its energy in the absence of damping. How many oscillations the tuning fork will make in this time.
7. The quality factor in a damped oscillator is $10e^{-4}$. After what time will its energy falls to $1/10$ of its initial value. The frequency of the oscillator is 200 Hz.
The quality factor of a spring loaded with 0.8 kg vibrating with frequency of vibration 10 Hz is 80. Calculate mechanical resistance.
8. A spring suspended from a rigid support carries a mass of 100g at its lower end and oscillates with a frequency of 10Hz. The amplitude reduces to half its initial value in 1minute. Calculate the damping constant and relaxation time.
9. The quality factor in a damped oscillator is $10e^{-4}$. After what time will its energy fall to $1/10$ of its initial value. The frequency of the oscillator is 200 Hz.
10. A pendulum oscillates 100 times in a second. The quality factor of the pendulum is 1000. Calculate the time in which the amplitude of the pendulum decays to $1/e^4$ of its initial value.